

# Shubhojyoti Datta Chaudhuri

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## SUMMARY

Ambitious software engineer and builder who shipped production REST APIs for a client in an Agile team, diagnosed and fixed a production timeout, and is independently building a fintech engine that reconciles credit card offers against bank transactions. Works mostly in Python and JavaScript with Java and Scala experience, practices code review and safe production changes, and uses AI tools while critically reviewing and validating every output. M.S. Computer Science, graduating February 2027.

## EDUCATION

**University at Buffalo, The State University of New York**  
*Master of Science in Computer Science, AI/ML Track | GPA: 3.7/4.0*  
**Chandigarh University**  
*Bachelor of Engineering in Computer Science | GPA: 3.2/4.0*

Buffalo, NY  
Aug 2025 – Feb 2027  
India  
Aug 2021 – May 2025

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, TypeScript, Scala, SQL, C, C++, Swift, HTML/CSS  
**AI & Machine Learning:** PyTorch, Hugging Face Transformers, NLP, LLMs, Model Evaluation, OpenCV, Tesseract, NumPy  
**Full-Stack & APIs:** Django, Django REST Framework, FastAPI, Flask, React, Next.js, REST APIs, Plaid API, Wix API  
**Cloud & Databases:** AWS (S3, Lambda, RDS), Docker, PostgreSQL, MySQL, SQLite, Redis, NoSQL  
**Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Concurrency, Database Design  
**Tools & Practices:** Git, GitHub, Code Reviews, AI-Assisted Development, pytest, Unit Testing, Linux, Agile

## EXPERIENCE

### LookThePart

*Software Developer*

Remote

Jan 2026 – Jun 2026

- Shipped 10+ new RESTful API endpoints and maintained 50+ more in Python with Django REST Framework for authentication, product search, post sharing, and media handling in an Agile team using Git code reviews.
- Optimized product search in the Swift (UIKit) iOS app, improving autocomplete suggestions and search results backed by the Django REST product search API via Alamofire.
- Modeled 15+ relational database entities in PostgreSQL with the Django ORM across entertainment, character, product, post, and media domains.
- Engineered Pillow-based image processing and thumbnail generation in a client-owned AWS S3 media pipeline, handling upload and optimized cloud delivery for 10,000+ image assets.
- Integrated the Wix Blog API to auto-generate blog drafts, then diagnosed a production 504 timeout caused by uploading 40 images per draft one by one and resolved it by parallelizing the uploads.

### VertexPlus Technologies Limited

*Software Engineer Intern*

India

Jun 2024 – May 2025

- Co-developed a window-sticker OCR pipeline in Python with OpenCV and Tesseract for a US automotive platform, extracting VIN, specs, and features into structured JSON for a Django REST API and AWS S3.
- Implemented layout detection with OpenCV template matching against 33 manufacturer logo templates and a 0.8 confidence threshold, routing each image to one of 12 layout-specific parsers covering 18+ vehicle brands.
- Built OCR image preprocessing: fixed-resolution resizing, grayscale conversion, noise cleanup, region cropping, and tiling.
- Co-designed the Vehicle IQ collision-prevention scoring algorithm, weighting normalized safety features with parent/child feature expansion and scoring against both dealer data and the NHTSA vPIC VIN decoder.

## PROJECTS

### Tally – Fintech Reconciliation Engine (In Progress) | Python, Flask, JavaScript, Plaid API

Aug 2026 – Present

- Independently building a full-stack Flask app that verifies whether credit card cashback offers (Amex, Chase) posted, using a date-windowed matching engine that classifies each offer as Posted, Awaiting, Missing, or Pending.
- Engineered merchant normalization and fuzzy matching that strips 20+ payment-processor prefixes and store codes from bank descriptors, and ranked issuers with a 95% Wilson score confidence-interval reliability index.
- Developed a REST API and JavaScript dashboard that generates claim letters for missing credits, with 16 pytest unit tests; now integrating the Plaid API for live transaction sync and an LLM pipeline that parses offer emails into JSON.

### Sentiment & Emotion Analyzer – Transformer NLP | PyTorch, Hugging Face, Flask, React

Jan 2026 – May 2026

- Fine-tuned five transformer models (DistilBERT, BERT, RoBERTa, XLNet, BERTweet) across 1.6M+ tweets with PyTorch and Hugging Face; BERTweet led at 86.4% accuracy and 82.3 F1, with McNemar tests confirming significance ( $p < 0.05$ ).
- Built an 85.4% majority-vote ensemble and a 28-label GoEmotions classifier (0.57 micro-F1), and served all models through a Flask REST API and React frontend.

### PintOS Kernel & Scheduler | C, Operating Systems

Aug 2025 – Dec 2025

- Implemented a Multi-Level Feedback Queue Scheduler (MLFQS) with 17.14 fixed-point recent-cpu and load-avg tracking.
- Built nested priority donation to resolve priority inversion and an interrupt-driven sleep timer to eliminate busy-waiting.